The University of Sydney

MOOC Introduction to Calculus

Notes for 'The escape velocity of a rocket'

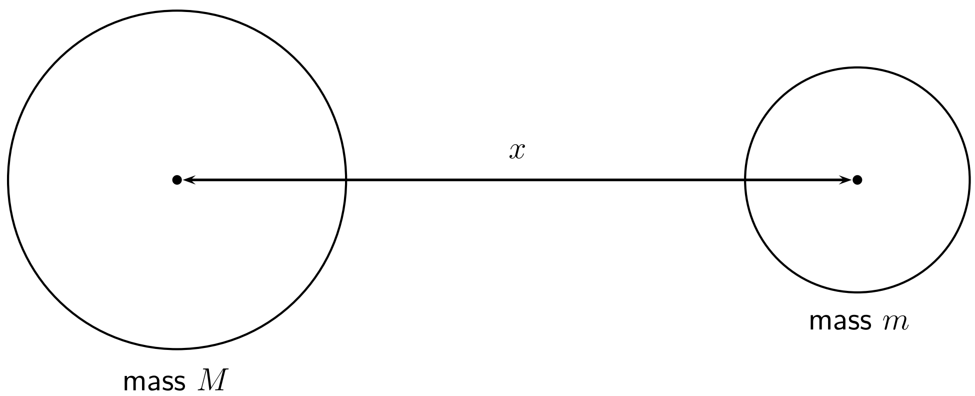
We aim to reconstruct, at least in spirit, Isaac's Newton's thought processes in estimating the *escape velocity* of an object, that is, the velocity one needs to impart to an object for it to escape the earth's gravitational field. This turns out to be independent of the mass of the object.

The term 'escape velocity' is often associated with rockets, but of course, modern rockets have a continuous supply of fuel. Newton was in fact thinking in terms of imparting enough energy to a cannonball, so that it can, in principle, escape from the earth. In the following development, we have the benefit of hindsight, and use Leibniz notation and the Riemann integral, neither of which were available to Newton, who had his own heuristic and other devices for conceiving the underlying mathematics.

The main idea is that the gravitational force $F = F(x)$ between two objects, which is a function of the distance x units between them, is proportional to their masses, say M and m respectively, and inversely proportional to the square of the distance x. Hence, we have

$$F(x) \;=\; \frac{GMm}{x^2}\;,$$

for some constant G (known as the *universal gravitational constant*). In the diagram below, the objects are perfectly spherical, but they could be irregular. The distance x is measured between their respective centres of gravity, which, in the case of spheres, becomes their usual centres.

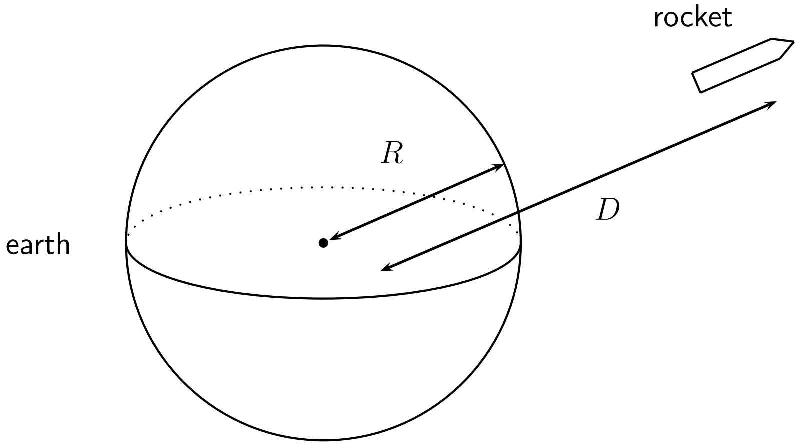


The numerator of the fraction in the expression for $F(x)$ makes sense, as the larger the masses, the greater one expects the gravitational force to be between them.

It also makes sense that $F(x)$ should be inversely proportional to the square of the distance. Remember, we showed that the surface area of a sphere is 4π times the square of the radius, so one can imagine the gravitational force radiating uniformly outwards from a spherical body, distributed evenly over 4π times the square of the distance from the centre. The gravitational force on any unit of area is therefore proportional to the inverse of the total surface area of an expanding sphere, which, in turn, is proportional to the inverse of the square of the distance to the centre of the sphere.

In fact, F becomes a function of several variables, if you include the masses of the objects as extra variables, though, in what follows, we think of M and m as constants, so that $F = F(x)$ becomes a function of the single variable x.

Denote the radius of the earth by R. Consider a rocket, moving away from the earth, and located in space, say, D units from the centre of the earth. We can ask how much energy $E = E(D)$, as a function of D, was expended, getting the rocket from the surface of the earth to its present position in space.



This can be described by the following definite integral:

$$E \;=\; E(D) \;=\; \int_R^D F(x)\, dx \;,$$

which we briefly explain. From elementary physics, if an object is moved a small distance, say Δx by a force $F(x)$, then the work done by the force, equivalently the energy expended, is calculated by multiplying the force by the distance, that is

$$F(x) \times \Delta x \;.$$

If we imagine the interval $[R, D]$ partitioned into tiny pieces,

$$x_0 = R \;<\; x_1 \;<\; x_2 \;<\; \ldots \;<\; x_{n-1} \;<\; x_n = D$$

and put $\Delta x_i = x_i - x_{i-1}$ for $i = 1$ to n, then an approximation of the work done, to get the rocket to its present position, is the Riemann sum

$$F(x_1)\Delta x_1 + F(x_2)\Delta x_2 + \ldots + F(x_n)\Delta x_n \;=\; \sum_{i=1}^{n} F(x_i)\Delta x_i \;.$$

By imagining this partition getting finer and finer, the approximation approaches the exact amount of work done, which then becomes the definite integral above. We can find this using the formula for $F = F(x)$, where m is the mass of the rocket and M is the mass of the earth, applying the Fundamental Theorem of Calculus:

$$
\begin{aligned}
E &= E(D) = \int_R^D F(x)\,dx = \int_R^D \frac{GMm}{x^2}\,dx = GMm\int_R^D \frac{dx}{x^2} \\
&= GMm\int_R^D x^{-2}\,dx = GMm\big[-x^{-1}\big]_R^D = GMm\Big(-D^{-1}-\big(-R^{-1}\big)\Big) \\
&= GMm\left(\frac{1}{R}-\frac{1}{D}\right).
\end{aligned}
$$

Imagine now the rocket travelling arbitrarily far way from the earth (which vanishes to a single point from the view of the rocket), that is, $D \to \infty$, so that $\frac{1}{D} \to 0$. We then get the following limit:

$$
\lim_{D\to\infty} E(D) = \lim_{D\to\infty} GMm\left(\frac{1}{R}-\frac{1}{D}\right) = GMm\left(\frac{1}{R}-0\right) = \frac{GMm}{R},
$$

which represents the amount of energy required for the rocket to escape the earth's gravitational field completely. Newton was in fact thinking about cannonballs, not rockets, and this calculation tells us that $\frac{GMm}{R}$ is the energy required to impart to a cannonball, when fired from the earth's surface, so that it manages to escape from the earth, where now m denotes the mass of the cannonball.

Newton wanted to estimate the initial velocity v that would need to be imparted to the cannonball for this to occur, and we call v the *escape velocity.*

From elementary physics, the energy of an object of mass m and velocity v is given by the formula $\frac{1}{2}mv^2$. Newton therefore wanted v such that

$$
\frac{mv^2}{2} = \frac{GMm}{R}.
$$

Cancelling m from both sides, and rearranging, we get $v^2 = \frac{2GM}{R}$, so that, taking the positive square root,

$$
v = \sqrt{\frac{2GM}{R}}.
$$

To turn this into an actual figure, observe that

$$
v = \sqrt{2R} \times \sqrt{\frac{GM}{R^2}}.
$$

The diameter of the earth is $2R$, which was calculated to be approximately 12.74×10^6 metres, by the Greek scholar and mathematician Eratosthenes, from about 200 BC, using clever measurements and geometry, and this was known to Newton. Further, at the surface of the earth, the gravitational force (in appropriate units) acting on an object of mass m is

$$
F(R) = \frac{GMm}{R^2} = m\left(\frac{GM}{R^2}\right).
$$

But force equals the mass m times the acceleration $\frac{GM}{R^2}$, for an object falling freely near the earth's surface. Newton knew this was about 9.8 m/sec^2. Hence, we may take the numerical value of $\frac{GM}{R^2}$ to be about 9.8, and so the estimate of the escape velocity becomes

$$
v = \sqrt{\frac{2GM}{R}} \approx \sqrt{12.74 \times 10^6} \times \sqrt{9.8} \approx 11{,}000 \text{ m/sec}.
$$

Thus the escape velocity of any object is about 11 kilometres per second.